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ISP Training Program – Edge Agent

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Agenda

- Brief Introduction to Advantech Solution Pilot
- Overview of Edge Agent
- Introduction to Edge Agent GitHub
- Quick Demo Projects
  - Factory Outfit Detection
  - Clearance Space Detection
- Exercises

Course Material: https://github.com/chunyu1021/2026-ISP-Global-Training-Program

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Brief Introduction to Advantech Solution Pilot

Advantech Solution Pilot

- Formerly knows as Innovation Solution Pilot (ISP)
- Five active software projects in ISP
- Based on NVIDIA's software solutions
- Added with Advantech customized features and integrations
- Aims for facilitating PoC process and development
- Further engineering IS NEEDED
- Works on ***Advantech MIC-AI Series***

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ISP Training Program

Training Date	Projects	Hardware	NVIDIA Base	GitHub
2026.04.15 (en) 2026.04.22 (zh)	Edge Agent	MIC-733	Agent Studio	LINK
TBD	VSS on Thor	MIC-742	VSS 2.4.0	LINK
TBD	Perceptor/ReMEmbR	MIC-732	Issac Perceptor/ReMEmbR	Perceptor/ReMEmbR
TBD	AMR SLAM Navigator	MIC-732	-	LINK
TBD	Omniverse/Issac Sim	-	-	-

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Overview of Edge Agent

Edge Agent Project Profile

- Originated from NVIDIA AI Lab's Agent Studio
- Additional features from Advantech's engineering
- Works on ***MIC-733-AO6A1 (64GB)*** + ***JetPack6.0***
- Features:
  - No code platform for building AI workflow
  - Easy access to ViTs, VLMs, and TTS models
  - NanoDB for RAG-powered model enhancement
  - User-driven development of new functions
  - Preset projects for demos

Edge Agent User Interface

No Code AI Workflow

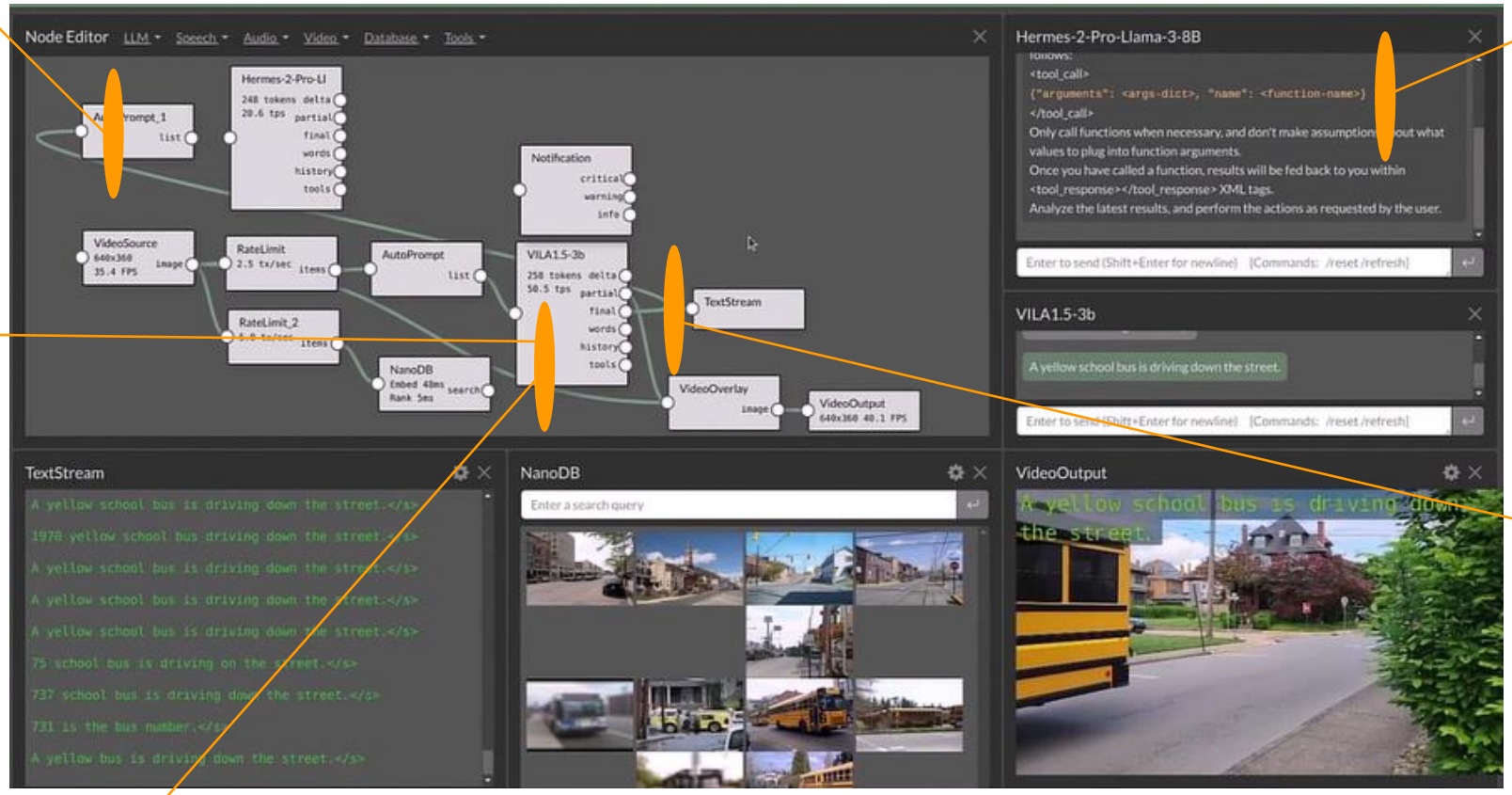
- Drag-and-drop function nodes

LLMs/VLMs/ NanoDB

- Preset models fetching directories
- Download the models with a click

Low Code Adjustments

- Flexible parameter adjustments for models



Text Stream /Piper TTS

- Text/audio interactions with models

Connect Node to Node

- Connect Nodes to activate AI workflow

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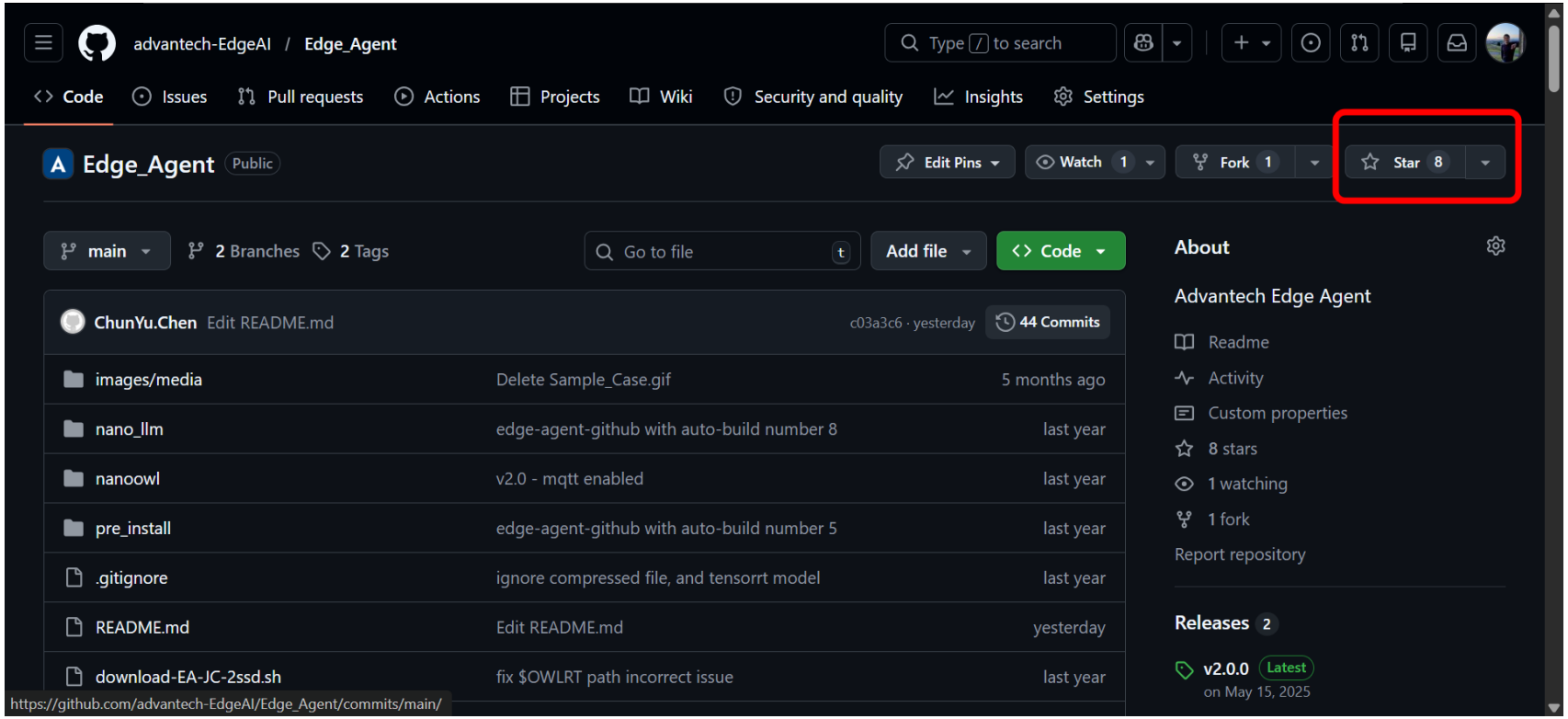
Introduction to Edge Agent GitHub

Edge Agent GitHub

- Link: https://github.com/advantech-EdgeAI/edge_agent
- ***Star the repo*** to get project updates
- Use ***project's wiki*** as your reference



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Quick Demo Projects

Factory Outfit Detection

- Project's wiki page
- Features:
  - Simple but used most project
  - Uses ViT model with Open Vocabulary Detection (OVD)
  - Customized object detections
  - (For advanced users) Customized functions – RecDetRes node

Clearance Space Detection

- Project's wiki page
- Features
  - Uses VLM with reliable prompt template
  - Prompt template is enhanced with In-Context Learning
  - Automatically saves images for specific conditions

AI Tools in Edge Agent

- Supported models:
  - Vision Language Models (VLM, e.g., LLaVA, VILA):
    - Describing images verbally or ***answering questions about visual content***
  - Vision Transformers (ViT, e.g., OWL-ViT) + Open Vocabulary Model
    - Open Vocabulary Model understands common vocabularies without training
    - Analyzing features within an image itself for tasks like ***object recognition***
- Add your own data: NanoDB + RAG
  - NanoDB: a vector database often used with VLMs for RAG
  - Retrieval Augmented Generation (RAG): a process that helps AI models “looks thing up” before they answer.
  - Try out the project: Adaptive Door Status Detection

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Exercises

Try Out Yourself!

- Run the Adaptive Door Status Detection project
  - Understand how to collect the data by using NanoDB
  - Understand how to use RAG to enhance model performance
  - Resource: How to add data to NanoDB
  - Resource: How to use NanoDB and RAG
- For advanced users, try to practice OpenWord FineTune project
- ***Build your own projects and share with us on GitHub Discussions***
- Any questions? Go to the project's GitHub Issues

Ask Questions on GitHub

- We use GitHub Issues to track questions.
- Before creating an issue…
  - You must sign in your GitHub account.
  - Try to find the answer in FAQ first.
- When you are creating a new issue…
  - Specify which BSP you are using.
  - Describe your question with details.
  - Always consider to upload supportive materials (e.g. screenshots, logs, etc.).
- After you submit an issue…
  - We will tag your issue with a label (e.g. bug, enhancement, question, etc.).
  - We will reply to your issue ASAP (you will be notified automatically by email).

Co-Creating the Future of the IoT World

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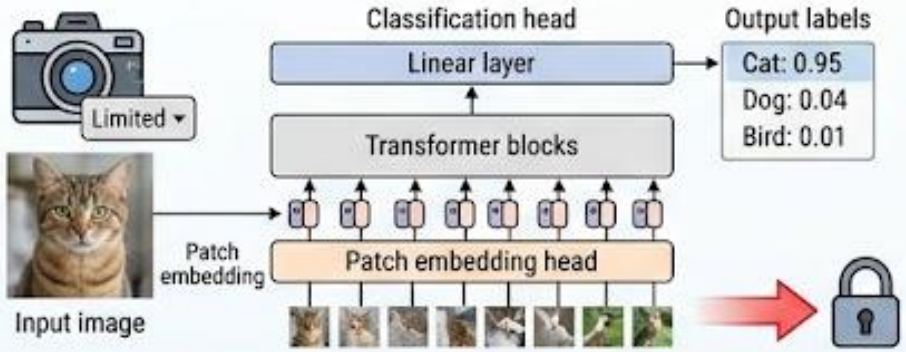
For Your References

Comparing Vision Transformer (ViT) and Open Vocabulary Models

Vision Transformer (ViT)

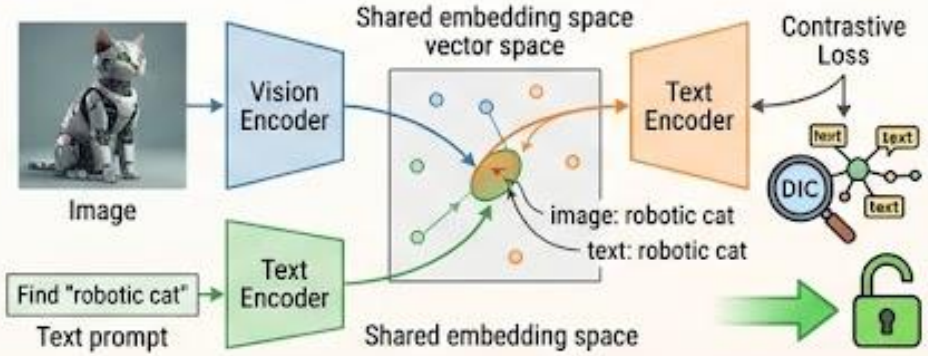
Fixed Vocabulary (Closed-Set)



- Standard ViT model (e.g., ImageNet)
- Fixed classification head
- Limited to a pre-defined set of categories
- Cannot recognize novel objects outside training data

Open Vocabulary Models (e.g., CLIP)

Flexible Language Alignment



- Uses Text Encoder (e.g., CLIP, OWL-ViT)
- Aligns images and text (Shared Space)
- Recognizes novel objects using language
- Generalizes to unseen concepts at inference

Source: Academic Research on Vision-Language Models

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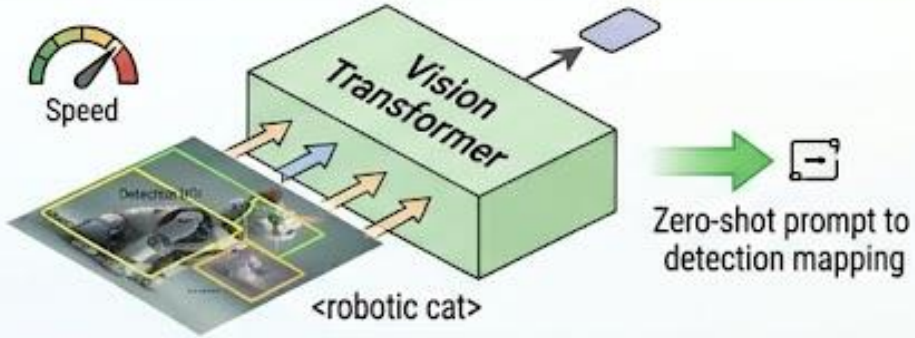
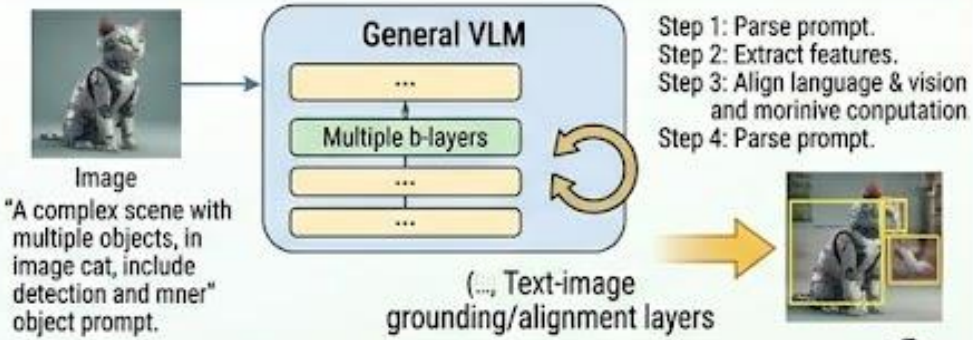
Why open-vocabulary ViT is efficient for object detection compared to general VLM (e.g. VILA)

A. open-vocabulary ViT (e.g., OWL-ViT)



- Lightweight architecture optimized for visual features
- Direct detection head (regression/classification) for objects
- Single-pass inference for detection
- Significantly lower computational overhead
- Scales efficiently for real-time applications

B. General VLM (e.g., VILA-like)



- Heavy model architecture integrating large LLM
- Requires multiple visual-text alignment steps per detection
- Multi-stage inference process increases latency
- High computational cost per image
- Inefficient scaling for dense object detection

* Academic Research

Summary | OPEN-VOCABULARY ViT OFFERS DIRECT MAPPING AND LOW LATENCY, OPTIMIZING RESOURCE USE FOR DENSE OBJECT DETECTION, WHERE GENERAL VLMs INCUR SUBSTANTIAL OVERHEAD FOR COMPREHENSIVE REASONING.

Source: Comparative Analysis of Specialized vs. General Vision Models

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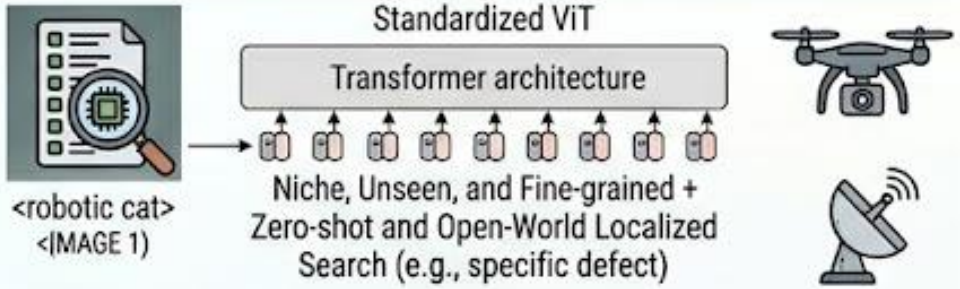
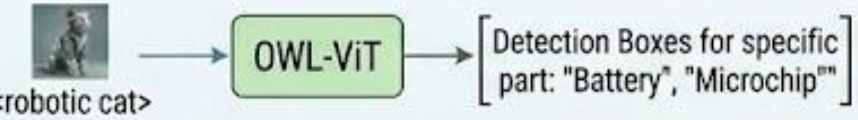
BEST USE CASES: OPEN-VOCABULARY ViT VS. GENERAL VLM (VILA-LIKE)

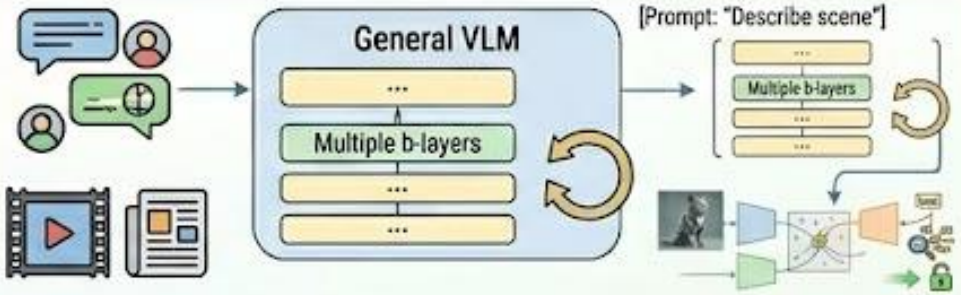
A. Open-Vocabulary ViT (e.g., OWL-ViT)



- Edge and On-Device Real-time Applications
- High-Speed and High-Volume Inspection (e.g., manufacturing, warehouse)
- Large-Scale Visual Data Analysis and Search



B. General VLM (e.g., VILA-like)



- Deep Multi-Modal Reasoning and Context Understanding
- Iterative, Multi-Stage VQA and Narrative Visual Tasks
- Complex Text-to-Image and Image-to-Text Grounding
- Broad Knowledge Transfer and General Problem Solving
- Offline or Batch Processing where Comprehension is Prioritized

* Academic Research

Summary | OPEN-VOCABULARY ViT EXCELLS IN FOCUSED, HIGH-SPEED VISUAL LOCALIZATION FOR SPECIALIZED OBJECTS; GENERAL VLMS FOR COMPREHENSIVE MULTI-MODAL REASONING, CONTEXTUAL UNDERSTANDING, AND ITERATIVE PROCESSING OVER BROAD KNOWLEDGE.

Source: Academic Research on Vision-Language Models and Specialized Vision Systems

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Integrating NanoDB, RAG, & General VLM for Context-Aware Solutions

User Query/ Image Input

New, niche gadget (<IMAGE-1)

"What is this and how do I fix it?"

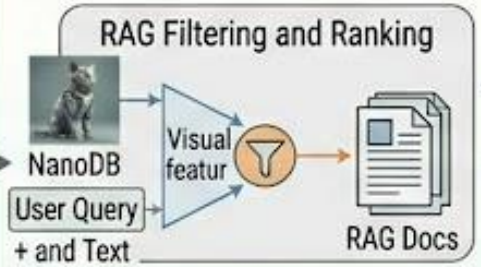
Enterprise Server

NanoDB: Local Fine-grained Visual-Text Database

Key	Value
...	...
...	...
...	...

- Stores specific product details, technical specs, and rare examples (like robotic components or defect types)
- Fast, direct semantic search for visual-text pairs
- > Fine-Grained localized visual information (from <-IMAGE-2)

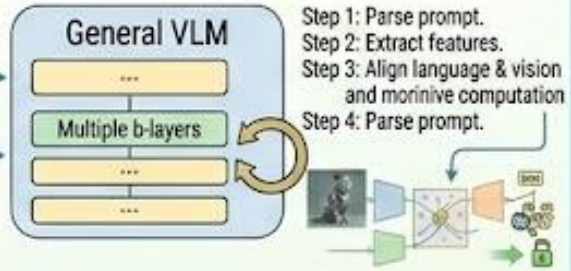
RAG Retrieval: Filtering, Ranking, & Aggregating Context



Ranked relevant documents/spec sheets (e.g., Component diagrams, troubleshooting guides)

- Synthesizes detailed local context for the LLM/VLM

General VLM (e.g., VILA-like): Context-Enhanced Reasoning & Generating



- Detailed and Accurate Context-Aware Response: "This is the X-1000 mainboard from [NanoDB source]. Step-by-step fix is in [RAG source]. Check component B3 for faults..."

* Academic Research

Summary INTEGRATING NanoDB (Niche Localized Context) WITH RAG (Aggregated Technical Data) EMPOWERS A GENERAL VLM TO PERFORM ACCURATE, FINE-GRAINED REASONING AND PROVIDE PRACTICAL, CORRECT ANSWERS BEYOND ITS PRE-TRAINED KNOWLEDGE.

Source: Concept for Combining Local Data, Retrieval Augmentation, & General VLM

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